

PatchSim: a modular simulation framework for patch-based metapopulation epidemiology

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Summary

PatchSim is an open-source Python framework for simulating infectious disease dynamics across spatially structured populations. It implements a patch-based metapopulation approach in which a landscape is partitioned into discrete geographical units (patches) — such as subdistricts, districts, or ecological zones — connected by a weighted contact network. Within each patch, disease progression is described by user-defined compartmental models (e.g., SIR, SEIR, SIRS) specified in a YAML configuration file, where each transition is written as source → target with a rate expression. PatchSim provides adaptive ordinary differential equation (ODE) and explicit-Euler solvers, Sobol sensitivity analysis via SALib (Herman & Usher, 2017), bounded least-squares calibration, and command-line and Python interfaces. Its configuration-first scope covers compartment-transfer models over supplied patch and group interaction data; scientific assumptions remain the modeller's responsibility. PatchSim is available on PyPI (`pip install patchsim`), with source code and documentation linked from the project record (Illikkal et al., 2026).

Statement of need

Mathematical modelling of infectious diseases is essential for understanding transmission dynamics, evaluating interventions, and informing public health policy (Keeling & Rohani, 2008). Metapopulation models — which capture the heterogeneity of disease spread across connected spatial units (Grenfell & Harwood, 1997) — are particularly valuable for diseases where population movement shapes outbreak trajectories, including foot-and-mouth disease in livestock (Guyver-Fletcher et al., 2025; Keeling et al., 2001; Tildesley et al., 2006), dengue in urban landscapes (Wesolowski et al., 2015), and respiratory infections across administrative regions (Balcan et al., 2009; Gatto et al., 2020).

Implementing such models commonly requires custom scientific code or a platform tied to a particular modelling abstraction. This adds software work to the scientific tasks of specifying transitions, preparing spatial inputs, and checking model assumptions.

PatchSim reduces this cost through a configuration-first design in which model structure — compartments, transitions, and parameters — is declared in a YAML file rather than implemented in code. The intended users are epidemiological modellers, researchers studying spatial disease dynamics, and public health analysts who need to rapidly prototype and compare scenarios across spatial configurations and disease systems. PatchSim was developed at ARTPARK, the AI and Robotics Technology Park at the Indian Institute of Science (IISc), to support active modelling work on livestock disease dynamics in India.

State of the field

Existing software spans several related abstractions: EpiFire (Hladish et al., 2012) and EpiModel (Jenness et al., 2018) address contact-network epidemiology; epipack (Maier, 2021) supports compartmental, stochastic, and network models; EMOD (Bershteyn et al., 2018) uses an individual-based architecture; and GLEaM (Balcan et al., 2010; Van den Broeck et al., 2011) represents global mobility-driven metapopulations.

Metapopulation frameworks closer to PatchSim include MetaWards (Woods et al., 2022), MEMilio (Bicker et al., 2026), SimInf (Widgren et al., 2019), MetaCast (Grunnill et al., 2024), and flepiMoP (Lemaitre et al., 2024). MetaWards is a stochastic metapopulation framework originally built for the electoral wards of Great Britain. MEMilio is a C++ core with Python bindings that offers graph-ODE and agent-based models. SimInf declares stochastic transitions as strings over nodes linked by scheduled livestock movements. MetaCast broadcasts a user-written Python ODE over subpopulations and bundles Latin hypercube sensitivity analysis. flepiMoP is an R and Python forecasting pipeline that declares compartments and transitions in YAML over a mobility matrix.

PatchSim is an independent implementation, but it shares both its name and its modelling lineage with an earlier package developed at the Network Systems Science and Advanced Computing division of the University of Virginia (Srini Venkatramanan et al., 2020). That package implements a metapopulation SEIR model and has been applied to seasonal influenza vaccine allocation (Srinivasan Venkatramanan et al., 2019) and to county-scale influenza forecasting in the United States (Srinivasan Venkatramanan et al., 2021). The earlier implementation was also used for age- and district-stratified COVID-19 vaccine allocation in Karnataka, as documented in a medRxiv preprint (Adiga et al., 2021). The approach was subsequently taken up in collaborative modelling work at the Indian Institute of Science and applied to foot-and-mouth disease in Indian livestock populations, which is the setting from which the present work grew. Where the earlier package provides a fixed SEIR compartmental structure serving a specific forecasting application, the framework described here generalises that approach: compartmental structure is treated as user-supplied configuration, so SIR, SEIR, SIRS, or arbitrary user-defined compartment sets are expressed in YAML without modifying source code. The name is retained to acknowledge this lineage. The earlier implementation is distributed as NSSAC/PatchSim; this package is distributed as patchsim from dsih-artpark/patchsim.

We did not build on the three closest tools for the following reasons. flepiMoP already declares compartments and transitions in YAML over a mobility matrix and offers a deterministic solver. At the time of writing it lacks variance-based sensitivity indices and bounded least-squares calibration, fits by Markov chain Monte Carlo, and its R and Python pipeline is larger than the single package the Karnataka work needed. MetaCast expresses the model as Python code, so a change in compartments is a code change. SimInf is stochastic only and moves animals by scheduled events rather than through a contact matrix, so it does not provide the deterministic comparison over district contact data that the Karnataka work needed.

PatchSim contributes the combination none of the three offers in one Python package with a command-line interface: a compartment graph and rate expressions declared in configuration, a user-supplied weighted patch matrix, categorical group stratification, a deterministic solver, Sobol sensitivity analysis, and bounded calibration. A modeller can switch between compartment structures, solvers, spatial networks, and group interaction inputs without rewriting the runtime.

Software design

PatchSim's central design principle is that compartmental model structure is declared in YAML rather than implemented in code. Transitions are written as source -> target keys, each paired with a rate formula:

```
compartments: [S, I, R]
Parameters: {beta: 0.2, gamma: 0.1}
Transitions:
  "S -> I": "beta"
  "I -> R": "gamma * I"
```

A rate expression that does not name its source compartment is multiplied by that compartment, so "S -> I": "beta" gives a local flow of βS_i . For focal patch i in an ungrouped multi-patch model, the infectious pressure is $\lambda_i(t) = \sum_j W_{ij} I_j(t) / N_j$, and the infection flow in this example is $\beta S_i \lambda_i$; a single ungrouped patch applies no coupling, and grouped models also multiply by the group interaction matrix. This is a residence-based coupling in the sense of the Lagrangian multi-patch models of Sattenspiel & Dietz (1995) and Citron et al. (2021). Patch populations stay fixed at N_i , and residents of patch i are exposed to the resident prevalence of each patch j in proportion to W_{ij} ; no population moves. The prevalence at patch j counts residents only, whereas the full Sattenspiel–Dietz model counts visiting infectives and visitors in both numerator and denominator.

A supplied W is applied as given and its rows are not normalized, so the scale of W multiplies β . The built-in contact-network generator can produce row-normalized weights with a self-share diagonal. Whether a supplied matrix should do the same, and whether it comes from a gravity kernel or observed movement data, is the modeller's choice. The spatial coupling is bound to the compartment names S, I, and E, and λ_i counts I alone.

ODE mode uses `scipy.integrate.odeint`, SciPy's interface to the LSODA solver, while discrete mode uses deterministic explicit Euler. Both evaluate the same derivative function; discrete results should be checked at successively smaller time steps. As a deterministic patch model, PatchSim does not represent stochastic fade-out or invasion probability in small patches (Ball et al., 2015; Riley, 2007). The current network is fixed from day zero. Patch populations, initial states, network weights, and optional group interactions are read from CSV files resolved relative to the configuration.

Validation checks required fields, identifiers, finite input values, population totals, matrix dimensions, transition endpoints, and an arithmetic-only expression language; these checks do not establish scientific validity. The JSON Schema is available for editors and external tooling. Reproducible analysis workflows provide seeded first- and total-order Sobol indices (Sobol', 2001) and bounded multi-start calibration with input hashes and diagnostics. Built-in SIR, SEIR, SIRS, and SIS templates and complete worked examples are included in the documentation. This bounded modularity is configuration- and API-based; adding a numerical solver still requires Python development. PatchSim is released under GPL-3.0.

Research impact statement

PatchSim is in current research use at ARTPARK for foot-and-mouth disease vaccination-scenario analysis across Karnataka districts, where it replaced a project-specific implementation. The workflow models cattle and buffalo populations stratified by age group and species. That analysis is unpublished at the time of writing, and no publication enabled by the package has yet appeared. Published spatial analysis of foot-and-mouth disease in India includes a state-level Bayesian space-time model of reported outbreaks, which estimated a roughly 50% lower outbreak risk in states covered by the vaccination programme (Gunasekera et al., 2022). The Karnataka analysis applies a mechanistic patch model to the between-district allocation question.

Near-term significance rests on what a third party can install, run, and check. The package has been developed in public since April 2025, with two tagged releases on PyPI and an archived release record (Illikkal et al., 2026). Continuous integration runs the automated test suite, linting, and a documentation build on every pull request and push to the main

branch. The tests cover configuration validation, the expression evaluator, both solvers, group stratification, the sensitivity and calibration workflows, and the command-line interface. The seeded sensitivity and calibration outputs described above carry input hashes and diagnostics, so a third party can re-run and check a study. The documentation on Read the Docs includes worked examples.

AI usage disclosure

GitHub Copilot, OpenAI Codex CLI (GPT-5), and Anthropic Claude Code assisted with code, tests, and documentation, and with language checks and paraphrasing of the authors' draft of this paper. CodeRabbit assisted with code review. Model versions were not consistently retained for historical work. The authors reviewed, edited, tested, and validated all assisted outputs and made the scientific and architectural decisions.

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